INSURE — Actuarial & Telematics Calculation Formulas

COMPLETE MATHEMATICAL SPECIFICATIONS WITH LATEX & WORKED RUPEE EXAMPLES

Team: **Techsquad** (SYNAPSE '26)
Live Platform: **insurevsb.vercel.app**

1. DRIVING RISK SCORING ENGINE (SCORINGSERVICE.TS)

Formula 1: 5-Factor Combined Driving Risk Score

`backend/src/services/scoringService.ts (Line 110)`

$$S_{\text{risk}} = \min\left(100, \max\left(0,\ 0.30R_{\text{speed}} + 0.25R_{\text{brake}} + 0.15R_{\text{accel}} + 0.15R_{\text{corner}} + 0.15R_{\text{distract}}\right)\right)$$

- S_{risk}: Driving Risk Score normalized between $[0, 100]$ (0 = ideal defensive driving, 100 = severe danger).
- R_{speed}: Speeding risk indicator (30% weight).
- R_{brake}: Harsh deceleration/braking risk indicator (25% weight).
- R_{accel}: Rapid acceleration risk indicator (15% weight).
- R_{corner}: Harsh cornering/lateral force indicator (15% weight).
- R_{distract}: Distracted driving/phone usage indicator (15% weight). Night driving is strictly excluded from risk scoring.

Worked Example: Aggressive Driver Session

Given telematics readings: $R_{\text{speed}} = 60.0$, $R_{\text{brake}} = 40.0$, $R_{\text{accel}} = 50.0$, $R_{\text{corner}} = 20.0$, $R_{\text{distract}} = 30.0$:

$$S_{\text{risk}} = (0.30 \times 60) + (0.25 \times 40) + (0.15 \times 50) + (0.15 \times 20) + (0.15 \times 30) = 18 + 10 + 7.5 + 3 + 4.5 = \mathbf{43.0}$$

Formula 2: Final Driving Safety Score (0–100 Scale)

`backend/src/services/scoringService.ts (Line 120)`

$$S_{\text{safety}} = \max\left(0,\ \min\left(100,\ \text{round}(100 - S_{\text{risk}})\right)\right)$$

- $S_{\text{safety}} \geq 80$: Classified as **Low Risk (Safe Driver)** → Qualifies for **10% Dynamic Discount**.
- $50 \leq S_{\text{safety}} < 80$: Classified as **Moderate Risk** → **0% Standard Baseline**.
- $S_{\text{safety}} < 50$: Classified as **High Risk** → Incurs **+20% Risk Surcharge**.

Worked Examples:

1. **Safe Commuter** ($S_{\text{risk}} = 0.0$): $S_{\text{safety}} = 100 - 0 = \mathbf{100/100}$ → Safe Driver (Qualifies for 10% Discount).
2. **Aggressive Commuter** ($S_{\text{risk}} = 43.0$): $S_{\text{safety}} = 100 - 43 = \mathbf{57/100}$ → Moderate Risk Baseline.

Formula 3: Speeding Risk Sub-Indicator

`backend/src/services/scoringService.ts (Line 84)`

$$R_{\text{speed}} = \max\left(0,\ \min\left(100,\ P_{\text{speeding}} \cdot 2.5 + N_{\text{severe}} \cdot 10\right)\right)$$

- P_{speeding}: Percentage of trip duration exceeding the $80\ \text{km/h}$ highway threshold.
- N_{severe}: Count of severe speeding violations (speed $> 100\ \text{km/h}$).

Worked Example:

If a vehicle spent 12% of trip duration above $80\ \text{km/h}$ and had 2 severe bursts over $100\ \text{km/h}$:

$$R_{\text{speed}} = \min(100,\ 12 \times 2.5 + 2 \times 10) = 30 + 20 = \mathbf{50.0/100}$$

Formula 4: Event Rate per 100 Kilometers Normalization

`backend/src/services/featureService.ts (Line 210)`

$$\text{Rate}_{\text{event}} = \left(\frac{N_{\text{events}}}{D_{\text{total}}}\right) \cdot 100$$

- N_{events}: Count of harsh braking ($< -4\ \text{m/s}^2$), rapid acceleration ($> 3.5\ \text{m/s}^2$), or cornering spikes.
- D_{total}: Total accumulated driving distance in kilometers (normalizes exposure so long commutes are fair).

Worked Example:

A commuter registers 3 harsh brake events across a $20\ \text{km}$ commute:

$$\text{Rate}_{\text{brake}} = \left(\frac{3}{20}\right) \times 100 = \mathbf{15.0\ events\ per\ 100\ km}, \quad R_{\text{brake}} = \min(100,\ 15.0 \times 5.0) = \mathbf{75.0/100}$$

2. FAIR DYNAMIC PREMIUM CALCULATION ENGINE (PRICINGSERVICE.TS)

Formula 5: Total Annual Indicative Premium `backend/src/services/pricingService.ts (Line 98-100)`

$$P_{\text{final}} = \max\left(0,\ P_{\text{base}} + A_{\text{behaviour}} + A_{\text{accident}}\right)$$

$$\text{where} \quad A_{\text{behaviour}} = P_{\text{base}} \cdot r_{\text{behaviour}}, \qquad A_{\text{accident}} = P_{\text{base}} \cdot r_{\text{accident}}$$

- P_{base}: Approved base annual premium (Configured at **₹2,000.00**).
- $r_{\text{behaviour}}$: Telematics behaviour adjustment rate:
  - $S_{\text{safety}} \geq 80 \implies r_{\text{behaviour}} = \mathbf{-0.10}$ (-10% discount → save ₹200).
  - $50 \leq S_{\text{safety}} < 80 \implies r_{\text{behaviour}} = \mathbf{0.00}$ (0% neutral baseline).
  - $S_{\text{safety}} < 50 \implies r_{\text{behaviour}} = \mathbf{+0.20}$ ($+20\%$ risk surcharge → add ₹400).
- r_{accident}: Independent accident history adjustment (Legally separated from driving score):
  - 0 accidents OR unverified/unknown fault incident: $r_{\text{accident}} = \mathbf{0.00}$.
  - 1st verified driver at-fault accident: $r_{\text{accident}} = \mathbf{+0.05}$ ($+5\%$ probationary adjustment → add ₹100).
  - 2+ verified driver at-fault accidents: $r_{\text{accident}} = \mathbf{+0.20}$ ($+20\%$ repeat fault surcharge → add ₹400).

Worked Example 1: Cautious EV Commuter (Ravi)

Base Premium = ₹2,000 | Safety Score = 100/100 | 0 Accidents on record:

$$A_{\text{behaviour}} = 2000 \times (-0.10) = -200, \quad A_{\text{accident}} = 2000 \times 0.00 = 0$$

Final Annual Premium: ₹2,000 − ₹200 + ₹0 = **₹1,800.00 / year (10.00% Dynamic Discount)**

Worked Example 2: Risky Driver with 1 Verified Collision (Rahul)

Base Premium = ₹2,000 | Safety Score = 35/100 | 1 Verified At-Fault Incident:

$$A_{\text{behaviour}} = 2000 \times (+0.20) = +400, \quad A_{\text{accident}} = 2000 \times (+0.05) = +100$$

Final Annual Premium: ₹2,000 + ₹400 + ₹100 = **₹2,500.00 / year (+25.00% Net Surcharge)**

3. IOT ACCELEROMETER & COLLISION ALERTING (SKETCH.INO & IOT.TS)

Formula 6: 3-Axis Euclidean Impact Force Vector Magnitude `ESP32 Firmware (sketch.ino) & backend/src/routes/iot.ts (Line 390)`

$$g_{\text{impact}} = \frac{\sqrt{a_x^2 + a_y^2 + a_z^2}}{g_0}$$

- a_x, a_y, a_z: Real-time linear acceleration axes sampled from MPU-6050 sensor at I2C address 0x68 in m/s^2.
- g_0: Gravitational constant ($9.80665\ \text{m/s}^2$).
- **Collision Trigger**: If $g_{\text{impact}} \geq 5.0\text{g}$ OR deceleration $\leq -12.0\ \text{m/s}^2$, creates a provisional collision alert with `fault_status = 'unknown'`.

Worked Examples: Nominal Cruising vs. Impact Crash Spike

1. **Normal Urban Cruising**: $a_x = 0.4$, $a_y = 0.3$, $a_z = 9.8\ \text{m/s}^2$:

$$g_{\text{impact}} = \frac{\sqrt{0.4^2 + 0.3^2 + 9.8^2}}{9.81} = \frac{9.81}{9.81} = \mathbf{1.00g} \quad (\checkmark\ \text{Nominal road baseline})$$

2. **Collision Crash Spike (Scenario C)**: Sudden impact shock $a_x = -68.0\ \text{m/s}^2$, $a_y = 0.0$, $a_z = 25.0\ \text{m/s}^2$:

$$g_{\text{impact}} = \frac{\sqrt{(-68.0)^2 + 0^2 + 25.0^2}}{9.81} = \frac{72.45}{9.81} = \mathbf{7.38g} \quad (\text{⚠ COLLISION ALERT TRIGGERED})$$

4. UNDERWRITING PROFITABILITY & PORTFOLIO ECONOMICS (PROFITABILITYSERVICE.TS)

Formula 7: Gross Earned Premium Revenue

`backend/src/services/profitabilityService.ts (Line 93)`

$$R_{\text{premium}} = N_{\text{customers}} \cdot \bar{P}_{\text{annual}}$$

Worked Example: INSURE Expected Adoption Preset

Portfolio of $N = 1,300$ policyholders with average earned premium $\bar{P} = 1,800$ INR:

$$R_{\text{premium}} = 1300 \times 1800 = \mathbf{23,40,000\ INR} \quad (₹23.40\ \text{Lakhs Gross Revenue})$$

Formula 8: Net Underwriting Profit

`backend/src/services/profitabilityService.ts (Line 97-99)`

$$\Pi_{\text{underwriting}} = R_{\text{premium}} - C_{\text{claims}} - \left(E_{\text{operating}} + E_{\text{acquisition}} + E_{\text{platform}} + E_{\text{other}}\right)$$

Worked Example (Baseline vs. INSURE Comparison):

• **Baseline (Before INSURE)**: Revenue = ₹20.00L, Claims = ₹12.00L, Expenses = ₹5.00L:

$$\Pi_{\text{baseline}} = 2000000 - 1200000 - 500000 = \mathbf{3,00,000\ INR} \quad (₹3.00\ \text{Lakhs})$$

• **With INSURE (Safe Driving Adoption)**: Revenue = ₹23.40L, Claims = ₹13.00L, Total Expenses = ₹5.50L:

$$\Pi_{\text{INSURE}} = 2340000 - 1300000 - 550000 = \mathbf{4,90,000\ INR} \quad (+63.3\%\ \text{Profit Expansion!})$$

Formula 9: Loss Ratio, Expense Ratio & Combined Ratio

`backend/src/services/profitabilityService.ts (Line 105-113)`

$$\text{Loss Ratio (LR)} = \left(\frac{C_{\text{claims}}}{R_{\text{premium}}}\right) \cdot 100\%, \qquad \text{Expense Ratio (ER)} = \left(\frac{E_{\text{total}}}{R_{\text{premium}}}\right) \cdot 100\%$$

$$\text{Combined Ratio (CR)} = \text{LR} + \text{ER}$$

- $\text{CR} < 100\%$: Underwriting operation is profitable.
- $\text{CR} > 100\%$: Underwriting deficit (insurer pays out more in claims and expenses than collected in premiums).

Worked Example: INSURE Portfolio Health

With Claims = ₹13.00L, Total Expenses = ₹5.50L, Premium = ₹23.40L:

$$\text{Loss Ratio} = \left(\frac{13,00,000}{23,40,000}\right) \times 100 = \mathbf{55.56\%}, \qquad \text{Expense Ratio} = \left(\frac{5,50,000}{23,40,000}\right) \times 100 = \mathbf{23.50\%}$$

$$\text{Combined Ratio} = 55.56\% + 23.50\% = \mathbf{79.06\%} \quad (\text{Highly Profitable Portfolio})$$

Formula 10: Break-Even Policyholders & Break-Even Premium

`backend/src/services/profitabilityService.ts (Line 115-121)`

$$N_{\text{break-even}} = \left\lceil \frac{C_{\text{claims}} + E_{\text{total}}}{\bar{P}_{\text{annual}}} \right\rceil, \qquad \bar{P}_{\text{break-even}} = \frac{C_{\text{claims}} + E_{\text{total}}}{N_{\text{customers}}}$$

Worked Example:

If Total Incurred Costs (Claims + Expenses) = ₹18,50,000 and Average Premium = ₹1,800:

$$N_{\text{break-even}} = \left\lceil \frac{18,50,000}{1,800} \right\rceil = \lceil 1027.77 \rceil = \mathbf{1,028\ Policyholders}$$

Acquiring any customer above $1,028$ generates net positive underwriting profit.